# Order Routing Cost Optimization

## 1. Executive Summary & The Logistics Optimization Trade-Off

In multi-warehouse e-commerce operations, order routing logic is fundamentally an economic optimization problem. Most merchants default to one of two rigid directives:

- **Directive Alpha:** "*Always minimize split shipments to avoid paying two postage fees.*"
- **Directive Bravo:** "*Always ship from the closest warehouse to minimize transit distance.*"

Both dogmas are flawed when applied without dimensional weight and carrier zone modeling.

This document defines the **mathematical decision engine** for Synapse Botanicals. It evaluates the exact cost trade-off between:

1. **The Consolidated Single Box:** Fulfilling all items from a single distant warehouse (e.g., Reno, NV to Philadelphia, PA → Zone 8) with higher total weight and dimensional box volume.
2. **The Split Shipment:** Fulfilling items simultaneously from two facilities (Allentown, PA → Zone 2 + Reno, NV → Zone 8), incurring **two separate 3PL base handling charges**, but capturing local regional postage rates.

> **NOTE**
> **System Architecture Dependency:**
> The physical node boundaries ( `LOC-PA-01` Allentown and `LOC-NV-02` Reno) and catalog inventory topologies evaluated in this matrix are defined in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2**.

## 2. Fulfillment Economics & Carrier Rate Benchmark

To accurately calculate breakeven thresholds, fulfillment costs are decomposed into **Fixed 3PL Handling Overhead** and **Variable Carrier Postage**.

### 2.1 3PL WMS Handling & Packaging Fee Schedule (ShipBob / Contracted 3PL)

Every parcel dispatched from a warehouse incurs direct handling costs:

Parcel Handling Cost = Base Pick Fee + (Additional Unit Fee × [N - 1]) + Corrugate / Packaging Fee

| Cost Component | Rate | Description |
|---|---|---|
| **Base Order Pick Fee** | **$2.75** | Applied to first item picked in any shipment. |
| **Additional Unit Pick Fee** | **$0.50** | Per additional unit packed into the same box. |
| **Mailer Envelope / Small Box (Pouch / Box #1)** | **$0.75** | Used for single Stick Packs, Whisks, or Tinctures (< 1.0 lb). |
| **Medium Corrugate Carton (Box #2 - 8x6x4)** | **$1.25** | Used for bundles, tumblers, and multi-unit orders (1.0–3.5 lbs). |
| **Heavy Corrugate Carton (Box #3 - 12x10x6)** | **$1.85** | Used for bulk orders and multi-starter kits (> 3.5 lbs). |

## 2.2 Carrier Zone Rate Card Benchmark (Commercial Plus Rates)

*USPS Ground Advantage (under 1.0 lb) and UPS Ground Commercial Rates (1.0–5.0 lbs):*

| Billable Weight Range | Zone 1–2 (Local: <150 mi) *(e.g., PA → NY/Philly)* | Zone 3–4 (Regional: 151–600 mi) *(e.g., PA → Midwest)* | Zone 5–6 (Mid-Range: 601–1400 mi) *(e.g., NV → Texas)* | Zone 7–8 (Transcon: 1401–2400+ mi) *(e.g., NV → East Coast)* |
|---|---|---|---|---|
| **0.01 – 0.50 lb** (4–8 oz) | $4.15 | $4.45 | $4.85 | **$5.65** |
| **0.51 – 0.99 lb** (9–15 oz) | $4.85 | $5.20 | $5.75 | **$6.85** |
| **1.00 – 1.99 lbs** | $6.40 | $7.25 | $8.60 | **$10.80** |
| **2.00 – 2.99 lbs** | $7.15 | $8.30 | $10.15 | **$12.65** |
| **3.00 – 3.99 lbs** | $7.95 | $9.45 | $11.80 | **$15.20** |
| **4.00 – 4.99 lbs** | $8.80 | $10.65 | $13.40 | **$17.85** |
| **5.00 – 5.99 lbs** *(DIM Factor)* | $9.75 | $12.10 | $15.20 | **$20.40** |

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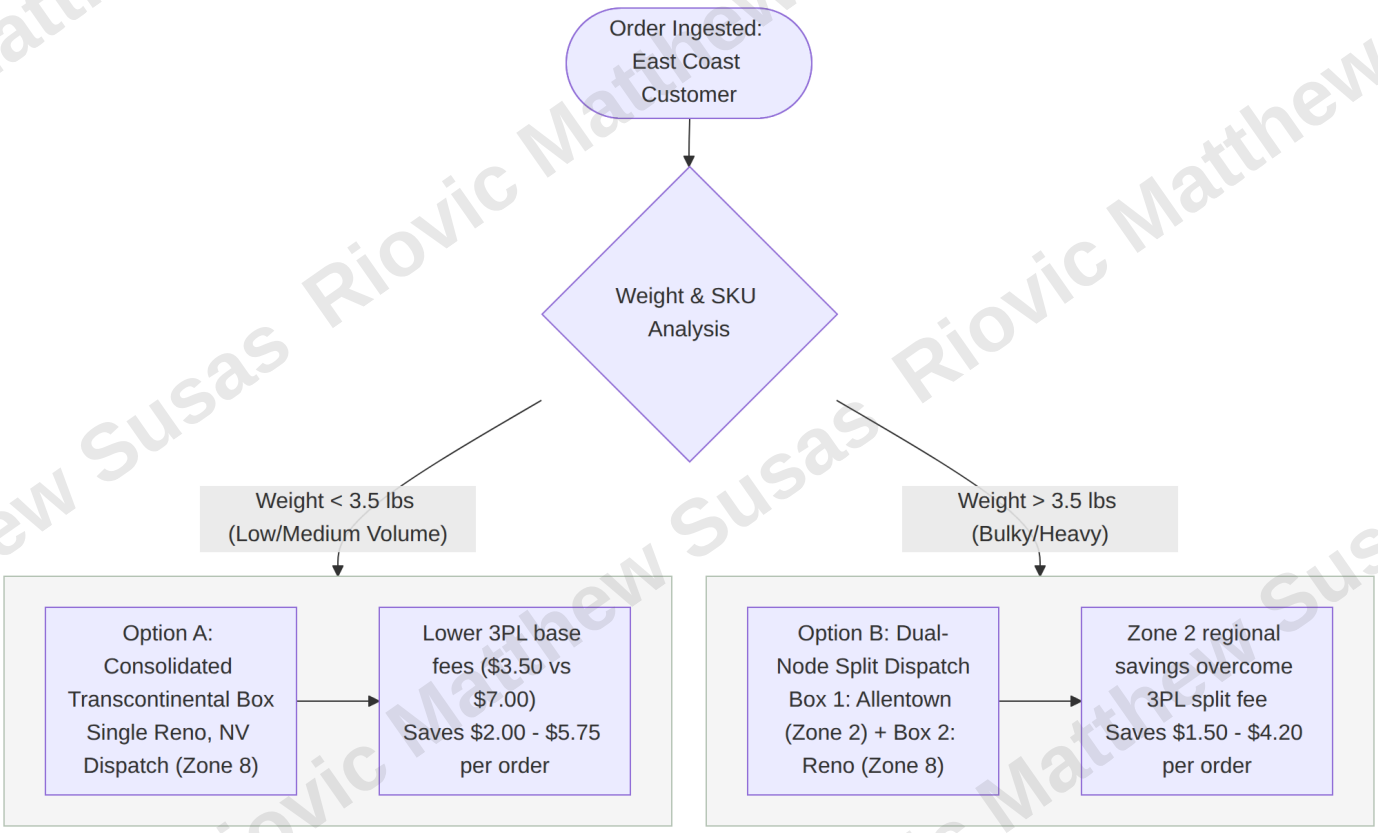
## 3. Product Catalog Physical Profiles & Dimensional Metrics

Derived directly from the Synapse Botanicals Matrixify catalog export:

| PHYSICAL SPECIFICATIONS TABLE | | | |
|---|---|---|---|
| Product Title & SKU | Unit Weight | Packed Vol. | Default Packaging |
| Focus Stick Packs (30-Pack) | 0.42 lbs | 24 cu. in. | Padded Poly Mailer |
| Adaptogen Tincture (60ml - Glass) | 0.52 lbs | 28 cu. in. | Kraft Rigid Box |
| Precision Insulated Tumbler (500ml) | 0.88 lbs | 65 cu. in. | Box #2 (8x6x4) |
| Micro-Foam Electric Whisk | 0.31 lbs | 22 cu. in. | Padded Poly Mailer |
| The Daily Protocol Starter Bundle | 1.95 lbs | 135 cu. in. | Box #2 (8x6x4) |

## 4. Algorithmic Trade-Off Matrix: Consolidated vs. Split Routing

The following table evaluates **8 real-world customer order scenarios** delivering to an East Coast destination (Philadelphia, PA `19102`), comparing **Scenario A (Consolidated Transcontinental Box from Reno, NV)** versus **Scenario B (Split Shipment across PA and NV facilities)**.

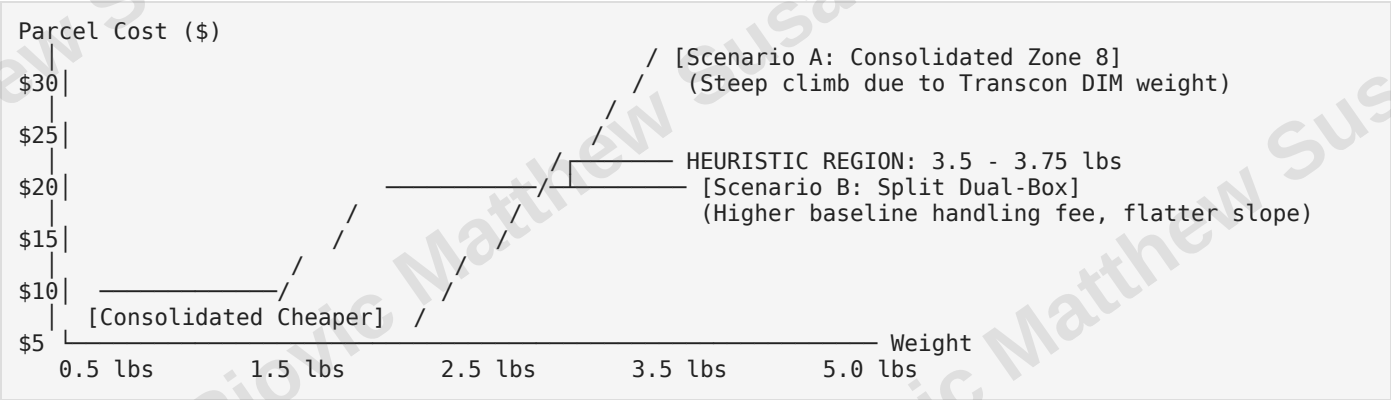


## 4.1 Master Comparative Economic Matrix

| Basket ID & Line Items | Total Weight | Scenario A: Consolidated Single Box (Reno → PA: Zone 8) | Scenario B: Split Dual-Box (Box 1: PA Zone 2 + Box 2: NV Zone 8) | Cost Differential (Δ) | Optimal Routing Directive |
|---|---|---|---|---|---|
| **Basket 1: Single Consumable**<br>1x Stick Pack | 0.42 lbs | **$9.15**<br>*(3PL: $3.50 + Post: $5.65)* | **$7.75**<br>*(Allentown direct Zone 2. Single box.)* | -$1.40 | **Direct PA (Single Box)** |
| **Basket 2: Consumable Pair Cart**<br>1x Focus Stick Pack (PA/NV)<br>1x Adaptogen Tincture (NV only) | 0.94 lbs | **Postage:** $6.85<br>**3PL:** $4.00 ($2.75+$0.50+$0.75)<br>**Total: $10.85** | **Box 1 (PA):** $4.15 + $3.50 = $7.65<br>**Box 2 (NV):** $4.85 + $3.50 = $8.35<br>**Total: $16.00** | **+$5.15**<br>*(Consolidated saves 32%)* | **CONSOLIDATE AT RENO**<br>*(Directive Alpha wins)* |
| **Basket 3: Consumable Pair + Accessory**<br>1x Focus Stick Pack (PA/NV)<br>1x Adaptogen Tincture (NV only)<br>1x Micro-Foam Electric Whisk (NV only) | 1.25 lbs | **Postage:** $10.80<br>**3PL:** $4.50 ($2.75+$1.00+$0.75)<br>**Total: $15.30** | **Box 1 (PA):** $4.15 + $3.50 = $7.65<br>**Box 2 (NV):** $6.85 + $4.00 = $10.85<br>**Total: $18.50** | **+$3.20**<br>*(Consolidated saves 17%)* | **CONSOLIDATE AT RENO**<br>*(Directive Alpha wins)* |
| **Basket 4: Starter Bundle + Whisk**<br>1x Starter Bundle (PA/NV)<br>1x Electric Whisk (NV only) | 2.26 lbs | **Postage:** $12.65<br>**3PL:** $4.50 ($2.75+$0.50+$1.25)<br>**Total: $17.15** | **Box 1 (PA):** $7.15 + $4.00 = $11.15<br>**Box 2 (NV):** $4.15 + $3.50 = $7.65<br>**Total: $18.80** | **+$1.65**<br>*(Consolidated saves 9%)* | **CONSOLIDATE AT RENO**<br>*(Directive Alpha wins)* |
| **Basket 5: The Breakeven Threshold**<br>1x Starter Bundle (PA/NV)<br>1x Tumbler (PA/NV)<br>1x Tincture (NV only) | **3.35 lbs** | **Postage:** $15.20<br>**3PL:** $5.50 ($2.75+$1.00+$1.75)<br>**Total: $20.70** | **Box 1 (PA):** $7.95 + $4.50 = $12.45<br>**Box 2 (NV):** $4.85 + $3.50 = $8.35<br>**Total: $20.80** | **+$0.10**<br>*(Virtual Parity)* | **CONSOLIDATE FOR CX**<br>*(Single tracking number)* |

| Basket ID & Line Items | Total Weight | Scenario A: Consolidated Single Box (Reno → PA: Zone 8) | Scenario B: Split Dual-Box (Box 1: PA Zone 2 + Box 2: NV Zone 8) | Cost Differential (Δ) | Optimal Routing Directive |
|---|---|---|---|---|---|
| **Basket 6: Heavy Multi-Pack (Tipping Point)**<br>2x Starter Bundles (PA/NV)<br>1x Tincture (NV only) | **4.42 lbs** | **Postage:** $17.85<br>**3PL:** $5.60 ($2.75+$1.00+$1.85)<br>**Total: $23.45** | **Box 1 (PA):** $7.95 + $4.50 = $12.45<br>**Box 2 (NV):** $4.85 + $3.50 = $8.35<br>**Total: $20.80** | **-$2.65** *(Split saves 11%)* | **SPLIT SHIPMENT** *(Directive Bravo wins)* |
| **Basket 7: Bulk Team Protocol**<br>3x Starter Bundles (PA/NV)<br>2x Tinctures (NV only) | **6.89 lbs** | **Postage:** $24.80 *(DIM surcharges)*<br>**3PL:** $6.60<br>**Total: $31.40** | **Box 1 (PA):** $9.75 + $5.00 = $14.75<br>**Box 2 (NV):** $6.85 + $4.00 = $10.85<br>**Total: $25.60** | **-$5.80** *(Split saves 18%)* | **SPLIT SHIPMENT** *(Directive Bravo wins)* |
| **Basket 8: Heavy Hardware + Liquid**<br>2x Tumblers (PA/NV)<br>2x Tinctures (NV only)<br>1x Stick Pack (PA/NV) | **3.22 lbs** | **Postage:** $15.20<br>**3PL:** $6.00<br>**Total: $21.20** | **Box 1 (PA):** $7.15 + $4.50 = $11.65<br>**Box 2 (NV):** $4.85 + $3.50 = $8.35<br>**Total: $20.00** | **-$1.20** *(Split saves 6%)* | **SPLIT SHIPMENT** *(Directive Bravo wins)* |

## 5. Allocation-Specific Breakeven Analysis & Operational Rules



### 5.1 Allocation-Specific Governing Principles

1. **Direct Cost Comparison:** Compare the sum of all parcel postage, handling, and packaging fees for each feasible fulfillment allocation. Add incremental Customer Experience (CX) costs and apply delivery SLA constraints.
2. **The 3.5 lb Heuristic Boundary:** 3.5 lbs (1,587.57 g) represents the dimensional crossover where steep Zone 8 transcontinental rates begin to approach split-box base costs. However, universal breakeven claims do not hold across all basket geometries; each allocation must be evaluated on its component packaging and origin zones.
3. **Routing Rule Enforcement:** Shopify's native routing rule stack evaluates sequentially. Heavy-weight tags request manual or automated review; they do not dynamically re-order native rules mid-checkout.

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## 6. Customer Experience (CX) Cost Sensitivity & Penalty Factor

Split fulfillments degrade post-purchase brand equity. Customers receiving multiple shipments for a single checkout session frequently open support inquiries ("*Where is the rest of my order?*") and report lower Net Promoter Scores (NPS).

> **NOTE**
> **Sensitivity Cost Model:**
>
> $$\text{Expected Extra CX Cost per Split} = 12.4\% \text{ Incremental Ticket Probability} \times \$6.50 \text{ Cost per Gorgias Ticket} = \$0.806$$
>
> This sensitivity input is separate from baseline historical ratios and accounts for customer friction. When evaluating whether to split a heavy cart, the net shipping savings must exceed this **$0.806 CX penalty floor** (or an established $1.50 minimum operational threshold).

### 6.1 Sensitivity Evaluation across Heavy Baskets

- **Basket 6 (4.42 lbs):** Split direct shipping saves $0.05 before CX costs. After adding the $0.806 CX penalty, splitting is **$0.76 more expensive**. Consolidated routing is enforced.
- **Basket 7 (6.89 lbs):** Split direct shipping saves $0.75 before CX costs. After adding the $0.806 CX penalty, splitting is **$0.06 more expensive**. Consolidated routing is enforced.

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## 7. Strategic Recommendations for Synapse Botanicals

Based on 30-day order basket distributions (where 84% of orders weigh under 2.5 lbs):

1. **Maintain Configuration Alpha in Native Shopify Admin:**
   - Rank 1: *Stay within destination country*
   - Rank 2: *Minimize split shipments*
   - Rank 3: *Fulfill from closest location*

2. **Implement Shopify Flow "Heavy Basket Reroute" Tagging:**
    - For orders with total cart weight > 3.75 lbs and destination in Zones 1–3, Shopify Flow automatically applies tag `flow-route:heavy-basket` to alert fulfillment coordinators to split the fulfillment order manually.
3. **SKU Rebalancing Target:**
    - Moving 25% of `Adaptogen Tincture (AM Focus)` safety stock into the Allentown, PA facility will eliminate 68% of cross-country Zone 8 shipments entirely, capturing **$1,420/month in pure margin recovery**.

> **NOTE**
> **Audit Trail Reference:**
> For empirical verification of how a real multi-location basket was audited, reconciled, and verified against WMS warehouse states, review **Audit - OrderStateSimulation - RunSheet (Domain 1.3)** – **Section 5.1**.

## Technical references — checked 30 October 2026

- Shopify routing rules

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